Earth Systems and Environment
https://doi.org/10.1007/s41748-024-00451-8

ORIGINAL ARTICLE

Influence of Protected Areas in the Carbon Cycle: A Machine Learning Approach for Future Forecasting

Pedro Lago-González[1] · Carolina Acuña Alonso[2,3] · Xana Álvarez[1]

Received: 10 June 2024 / Revised: 2 August 2024 / Accepted: 20 August 2024
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Abstract

Protected Areas (PAs) are destined to the conservation of natural resources, whose quantity and quality are menaced by Climate Change and anthropogenic pressure. The objective of this research was to quantify and forecast the Carbon Absorption of PAs, attending these factors. We used the Net Primary Productivity (NPP) and implemented machine learning algorithms, as Random Forests (RF), Gradient Boosting Trees (GBT), and Multilayer Perceptrons (MLPNN), to forecast it in four differentiated PAs of Galicia (NW Spain): the Central Massif, the Sil Canyons, Fragas do Eume, and the Tambre River. Two testing stages were carried, one in the areas where the model was trained, and another in the whole territory. Finally, we set several scenarios based on projections SSP2-4.5 and SSP 5-8.5 in and land use changes. GBT was the most accurate algorithm, with a Root Mean Squared Error (RMSE) of 0.05 kgC/m^2 (5.7% of the average NPP) and correlation of 0.9. RF obtained an error of 0.07 and correlation of 80%, and MLPNN 0.06 and 86%, respectively. The GBT obtained a RMSE of 0.04 kgC/m^2 and R^2 of 0.95 in the first test. These results were worsened in the second test, with an RMSE of 0.09 kgC/m^2 and 71% of correlation. For the SSP2-4.5 scenarios, a decrease around 7% can be expected, barely influenced by the land use. On the other hand, the SSP5-8.5 is expected to record a decrease of 5% in the NPP, with no significant differences between the land uses, but with a significant trend. All the pilot sites stick to this trend, except for the Tambre River, which was forecasted to record a slight increase of the NPP (~3%). The prediction and analysis of future scenarios can help the management of the territory, focusing on the mitigation of the effects of Climate Change in PAs.

Highlights

- The ecosystem carbon absorption can be modelled from meteorological data
- GBT models have proved to result in accurate predictions for this purpose
- This makes possible to forecast carbon absorption under future climatic scenarios
- The Carbon absorption in PAs of Galicia is projected to decline with CC
- Radiative forcing was found to favour NPP

✉ Pedro Lago-González
pedro.lago.gonzalez@uvigo.gal

Carolina Acuña Alonso
carolina.alonso@uvigo.es

Xana Álvarez
xaalvarez@uvigo.es

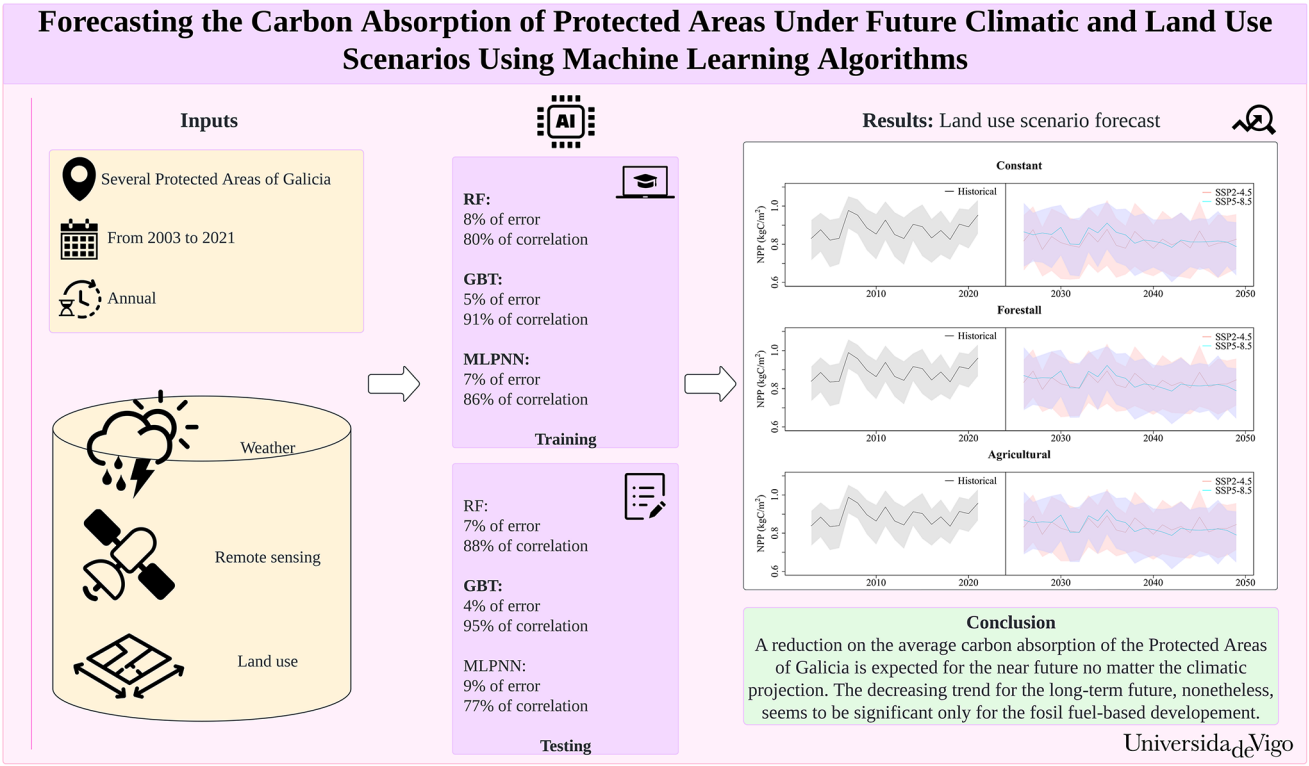
[1] Hydroforestry Geomodelling Group, School of forestry Engineering, University of Vigo, Campus A Xunqueira s/n, Pontevedra 36005, Spain

[2] CINTECX, Universidade de Vigo. Applied Geotechnologies Group Vigo, Vigo 36310, Spain

[3] Centro de Investigação e Tecnologias Agroambientais e Biológicas, Universidade de Trás-os- Montes e Alto Douro, Ap 1013, Vila Real 5001-801, Portugal

Published online: 09 September 2024

Graphical Abstract



Keywords Carbon absorption · Decision making · Ecosystem services · Natura 2000 network · Net primary productivity · Scenario projections

1 Introduction

Carbon Dioxide (CO_2) concentration in the atmosphere plays a well-known role as leading Climate Change factor (Taylor 2002). An anthropogenically driven augmentation on the CO_2 concentration on the atmosphere has led to an incrementation of the average superficial temperature (W. Zhong and Haigh 2013). In this issue, forest areas become an important mitigation agent, providing a quantity of ecosystem goods and services for climate regulation (Brown et al. 2007). Among others, some important ecosystem services supplied by these areas include mitigation of floods and droughts, microclimate regulation, preservation of biodiversity, or carbon storage (Daily 1997). The main risks for the quality and quantity of ecosystem services are the extreme weather events (Gauquelin et al. 2018), subject to Climate Change (Easterling et al. 2000); and anthropogenic pressure derived from land use (Gauquelin et al. 2018). These have been proved to increment the forest fire (Cary et al. 2006; Chas-Amil et al. 2013; Flannigan et al. 2016) and flooding risks (Ma et al. 2023; Pham and Alila 2024). In this sense, Protected Areas (PAs) are delimited spaces where the species biodiversity is to be conserved (Rodrigues et al. 2004). These is accomplished by limiting or banning the exploitation of natural resources (Lele et al. 2010), soothing the anthropogenic pressure they are exposed to. Nonetheless, the effectiveness of these areas on this aim is not enough when their management is inadequate (Rodrigues et al. 2004).

These areas cover around the 12% of the terrestrial land surface (Joppa et al. 2008), but they account around the 27% of the Greenhous Gasses (GHG) net forest sink, accounting for 47% in temperate forests (Harris et al. 2021). There seems, however, to exist a trend to the degradation of these areas in terms of forest extension, as they lost 4.1% of forested area between 2001 and 2018 (Wade et al. 2020). Henceforth, the SDG 17 (Sustainable Development Goal) advocates for the increase of the areas destined to this use (United Nations 2023), given their key role as a reservoir in the Carbon cycle (Duncanson et al. 2023). They act as so by fixating Carbon from the air and transforming it into organic matter (Johnson 2016), also known as Gross Primary Productivity (GPP). The difference between this process and plant respiration is the so-called Net Primary Productivity (NPP), which can be used as an estimation of the Carbon

fixed by plants (Wu et al. 2020). It depends directly and indirectly on the environmental conditions (Willmott et al. 1985) and, therefore, the Climate Change context in combination with the unsustainable land usage in natural areas (Xu et al. 2024) leads to a potential decrease in the ability of some areas to act as carbon storage.

Hence, it is of great interest to be able to model this service depending on meteorological variables and land use classification, in order to assess the potential effect of human activities under certain climatic scenarios. Machine learning has become a common tool for forecasting and in this issue Random Forests (RF), and Artificial Neural Networks (ANN), among others, have become a common approach for the prediction of the GPP (Zhu et al. 2024). RF is a widely known algorithm which consists of an ensemble of random decision trees (Breiman 2001). Bai et al. (2021) developed several machine learning models for the GPP prediction, RF among them, resulting in highly accurate results. On the other hand, Wei et al. (2017) tried joining the RF with the mechanistic Light-Use Efficiency Model (LUE), with a lesser accuracy than the RF alone. Similar to RF, Gradient Boosting Trees (GBT) are another ML algorithm consisting of an ensemble of simple decision trees, in which each tree is built to correct the residuals from the previous one (Friedman 2001). Bai et al. (2021) used two methods based in Gradient Boosting for GPP prediction: the Gradient Boosting Regression Trees (GBT) and the Light Gradient Boosting Machine (LightGBM), obtaining predictions that explained more than 95% of the data in both cases.

Regarding ANN, Multilayer Perceptron Neural Network (MLPNN) is a specific architecture of a feed-forward ANN designed specifically for data with monotonic relationships. It contains at least, one input, one output and one or more hidden layers, whose neurons are all connected to every single one of the following and the previous layer. This kind of algorithm was used for the prediction of CO_2 storage in terrestrial ecosystems in Melesse and Hanley (2005). Their best results were obtained for the wheat ecosystem and the worst for the grassland, with forest in the middle. Additional types of ANN have been applied to predict the Carbon absorption. Examples of these are the Long Short-Term Memory (Bai et al. 2021), and the Back Propagation Neural Network for the Net Ecosystem Exchange (NEE) prediction (Papale and Valentini 2003). Some other algorithms among the bibliography included Support Vector Machines for regression (SVM or SVR). Examples of this are Yang et al. (2007) and Ueyama et al. (2013). Both applied this algorithm for the prediction of GPP, in addition to the Ecosystem Respiration and the NEE for the latter. Yang et al. (2007) trained the model distinguishing between forest and non-forest areas.

These methods, however, have commonly been combined with remote sensing data, mostly relative to vegetation. These provide very accurate predictions but are not applicable to future scenarios where this kind of data is not available. Only recently some authors have developed models from data available for climatic projections (Gautam et al. 2022; Shi et al. 2024; Wang et al. 2024). These methods have proved their utility for the estimation of the role of forests as sink in the Carbon cycle under certain emission scenarios and how land use may interfere with these interactions. These studies, however, were carried in very large areas, but never in a reduced area with a high fragmentation rate of the land uses. Therefore, the aim of this paper is to apply an experimental, new methodology based on these techniques, specifically in the Autonomous Community of Galicia (NW Spain). The innovation of this methodology is due to 2 main factors: (i) the usage of data that allows the models to be applied to future climatic and land use scenarios; and (ii) the specific traits of the area of study, which is characterised for a highly fragmented land usage. Although PAs account for a total of 313,648 ha in this region (MITECO 2021), the potential effects of Climate Change in the Carbon storage under future climatic scenarios have not been widely studied in this specific region. Thus, this study is carried in a sample of four well differentiated interior PAs of this area, belonging to the Natura 2000 Network: Fragas do Eume, the Sil Canyon, the Central Massif and the Tambre River; all of which are inscribed as Sites of Community Importance (SCI) (EU, 2004) and as Special Areas of Conservation (SAC) (Consellería de Medio Ambiente Territorio e Infraestructuras 2014). Our approach is focused mostly on the influence that weather and climate factors produce on the NPP of forest and agricultural ecosystems.

Based on the bibliography best results, RF, GBT and ANN were proposed to address the aim of this study, as they showed the highest performance among bibliography. The models' performance and transferability were assessed and compared; several testing phases were carried with this purpose, one for the performance and the other for the transferability. The best performing model was implemented to scenarios based on near-future (2026–2050) emission and mitigation representative pathways (SSP2-4.5 and SSP5-8.5) and land use changes to assess the potential of the areas as a carbon reservoir. This approach represents an efficient and transferable methodology, based on information derived from a few meteorological variables which are accessible for future emission scenarios, easing its applicability in the field of natural resources management. This methodology was tested in several and well differentiated pilot sites, generally providing reliable results.

2 Materials and Methods

The proposed methodology consisted of collecting data of the weather, the land use and the NPP from several sources (details are included below), and harmonising data to a common spatiotemporal resolution. After, data was divided into two sets, the former for training a set of models to predict the NPP, and the latter for testing them. The model with best fitting combination of hyperparameters was used for the determination of the NPP in future scenarios, enabling us to carry out statistical comparisons between them, and to assess the influence of land use changes on the NPP, under several climatic pathways (Fig. 1).

2.1 Study Area

This study was carried in four different PAs of the Autonomous Community of Galicia (NW Spain) (Fig. 2.b). These were the Fragas do Eume (Fig. 2.d), the Sil Canyon (Fig. 2.e), the Tambre River (Fig. 2.c) and the Galician Central Massif (Fig. 2.f). The choice of these pilot sites was made for several reasons. Firstly, they are representative of the two predominant climate types of Galicia (Fig. 2.a) which are the full wet and the dry warm summer oceanic (Csb, Cfb). Besides, in the mountainous regions of the Central Massif, there are some small areas with full wet warm summer continental (Dfb) climate type (Wouters 2021). The mean annual temperature of this area in the period between 2002 and 2021 was 10.1 ± 0.3 ºC, which is significantly lower than in the other areas. The annual average of the Sil Canyon was 13.5 ± 0.3 ºC, in Fragas do Eume it was 12.7 ± 0.4 ºC, and 13.4 ± 0.3 ºC in the Tambre River. The mean annual accumulated precipitation was 1351 ± 250 mm, 1687 ± 317 mm, 1340 ± 232 mm and 1575 ± 286 for the Central Massif, the Sil Canyons, the Fragas do Eume and the Tambre River. The Sil Canyons and the Tambre River registered greater values of precipitation than the Central Massif and Fragas do Eume.

These areas latitudinally entail almost the whole territory of Galicia, signifying a representative sample of the variables more dependent on the latitude, as it is the downwelling radiation. Moreover, they are each at different altitudes which favours the representativeness of the samples of variables dependant on height, such as the temperature. Thus, the Central Massif represents the highest altitude among the sampling sites, with a maximum of 1776 m and an average height of 1166 m (Fig. 2.f). On their part, the Sil Canyon and Fragas do Eume altitudes are in average similar. The first is in average 406 m and the latter 410 m. They respectively range from 111 m to 847 m (Fig. 2.e) and from 4 m to 711 m (Fig. 2.d). On the other hand, the Tambre River altitude ranges in a lesser extent, from 201 m to 510 m, with an average height of 330 m (Fig. 2.c).

Other reason for the choice of the sites is that they are representative of different areas: the Fragas do Eume is a characteristic temperate rainforest on the climax stage; while the Sil Canyon is a gorge that has been widely used for the grape vines production. On the other hand, the Central Massif represents a natural mountainous area, with a huge variety of land uses. Lastly, the Tambre River is a waterway of rich ecological value that flows through some populated areas. The Spanish Information System of Land Cover (from its Spanish acronym, SIOSE) classification of the land use set the main usage of the areas of study as land areas not in other economic use. Specifically, this kind of area represents the 92% of the total land use of the Central Massif. In the remaining sampling sites, it is also the main use, but in a lesser extent, ranging from an 81% in the Sil Canyon, to a 48% in the Tambre River. The Fragas do Eume falls in the middle with a 63% of land destined for that kind of use. The second most observed land use was for agriculture. It represents the 6% of land uses in the Central Massif, the 12% in the Sil Canyons, the 22% in Fragas do Eume and 43% in the Tambre River. These two are the more observed land uses in the sampling sites in the year 2014 (last sampling year), however, some other include forestry, mining and quarrying, secondary production, community services, transport networks, residential use, and water areas not in other economic use. The Tambre River holds a remarkable variety of land use around its flow, whose district accounts for a 9.9% of the Wildland-Urban Interface (WUI) of the total in Galicia, making it susceptible of forest fires (Chas-Amil et al. 2013) and consequently to biodiversity loss (Fahrig 2003).

2.2 Dataset

Data of monthly maximum average and minimum average temperature and accumulated precipitation was retrieved from the WorldClim database (Fick and Hijmans 2017). From these, variables representative of the seasonal behaviour of the meteorology were derived. Relative humidity and dewpoint temperature were retrieved from the ERA5 reanalysis dataset (Hersbach et al. 2018; Muñoz Sabater 2019). Furthermore, data of monthly averaged downwelling shortwave radiation and latent and sensible heat fluxes was obtained from this source. It must be noted that these had a lower spatial resolution than the previous. On the other hand, data of the Net Primary Productivity (NPP) (Running and Zhao 2021) and a Digital Elevation Model (DEM) (JPL 2013) was provided by the Moderate-Resolution Imaging Spectroradiometer (MODIS) (AppEEARS Team 2024). The

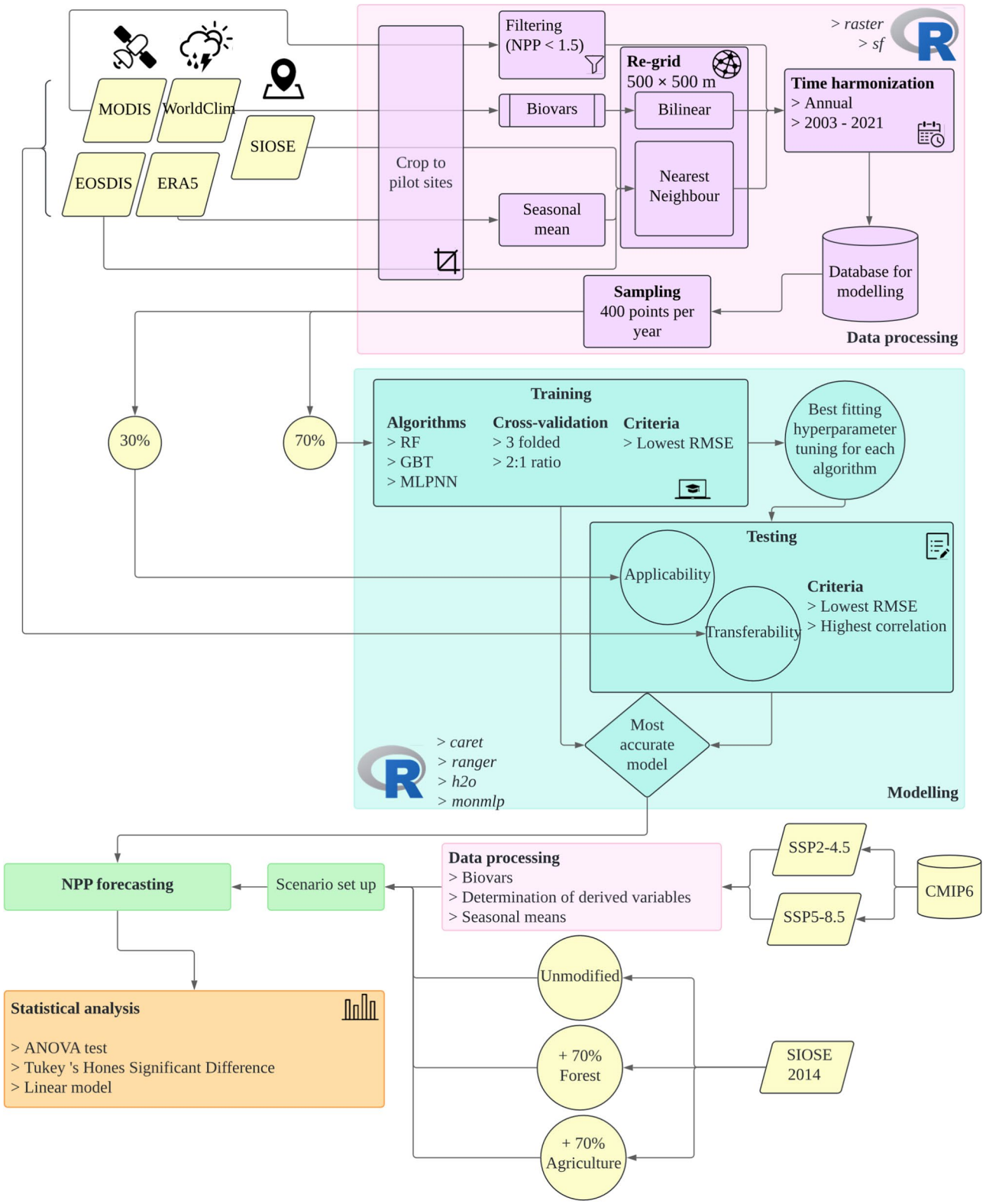


Fig. 1 Workflow diagram of the methodology used for the development, training, testing, application, and assessment of the models. MODIS: Moderate-Resolution Imaging Spectroradiometer; EOSDIS: Earth Observing System Data and Information System; ERA5: ECMWF (European Centre for Medium-Range Weather Forecasts) Reanalysis v5; SIOSE: Spanish Information System of Land Cover; NPP: Net Primary Productivity; RF: Random Forest, GBT: Gradient Boosting Trees; MLPNN: Multilayer Perceptron Neural Network; RMSE: Root Mean Squared Error; CMIP: Coupled Model Intercomparison Project; SSP: Shared Socioeconomic Pathway; ANOVA: Analysis of variance

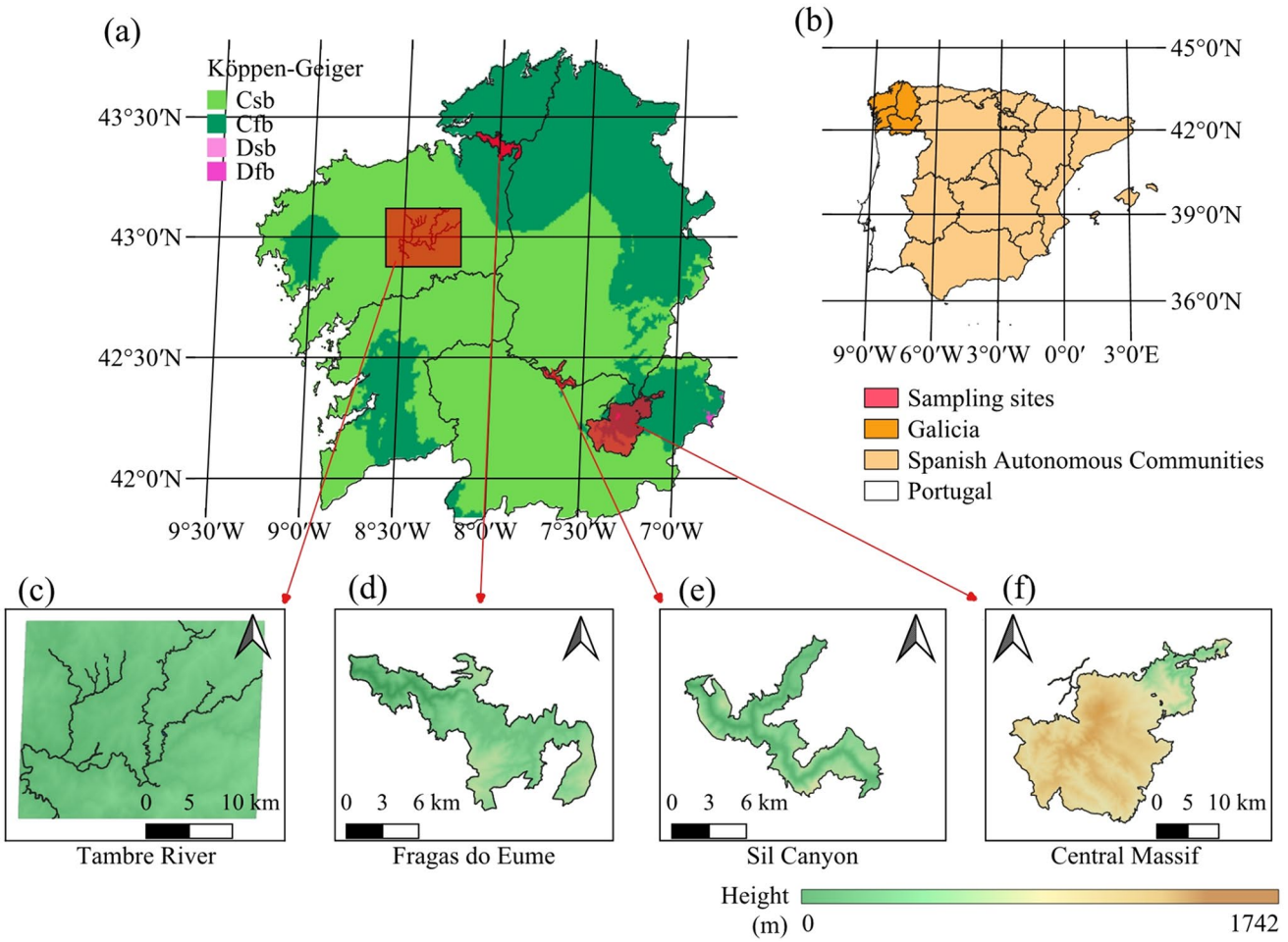


Fig. 2 Map of the study areas. (**a**) Climatic characterisation of Galicia with the sampling areas highlighted. (**b**) Location of Galicia in the Iberian Peninsula. (**c**), (**d**), (**e**), and (**f**) Digital Elevation Model of the sampling sites, respectively, the Tambre River, Fragas do Eume, the Sil Canyon and the Central Massif

classification of the Land Use was provided by the SIOSE (Instituto Geográfico Nacional 2014).

All the variables were resampled to a 500 m × 500 m grid and to annual temporal resolution. Data from WorldClim database was resampled through bilinear interpolation while data from other sources was interpolated through the nearest neighbour technique, as their resolution was too different from the target (ERA5, DEM) or given their classificatory nature (SIOSE) (Table 1). The R project packages *raster* and *sf* were used for the management of spatiotemporal data.

For the model to consider the effects of the seasonality of the meteorological and climatic variables, seasonal values of them were set as inputs. Furthermore, we considered the annual isothermality as an input variable as it has been linked to carbon storage (Estrada and Soares 2017; Rumman et al. 2018; Wang et al. 2023) and to species spread (Chen et al. 2023; Zhong et al. 2022) in other areas. The variables derived from maximum and minimum temperature, as well as precipitation, were calculated using the *dismo* library function *biovars*, we retained the variables 1–6 and 10–17 (see Hijmans et al. (2023) for further detail about the biovariables). On the other hand, data from ERA5 was simply seasonally averaged for each year.

2.3 Model Optimisation for the NPP Forecast

Several machine learning models were developed in the R project software environment. The *caret* library was used to set the models hyperparameter optimisation. Regarding the most widely used algorithms for this purpose, in this study we applied regression based in the Breiman RF (Breiman 2001) and the Monotonic MLPNN (H. Zhang and Zhang 1999) algorithms, based on the R libraries *ranger* and *monmlp*, respectively. We also tested the GBT (Friedman 2001), as it has been commonly used for the prediction of other ecosystem services, along with the RF (Almeida et al. 2024). The GBT used for this was based in the package *h2o*.

Table 1 Details on the original dataset

Variable	Units	Spatial res.	Temporal res.	Source
Mean max. temperature	ºC	2.5' × 2.5'	Monthly	World-Clim
Mean min. temperature	ºC	2.5' × 2.5'	Monthly	World-Clim
Accumulated precipitation	mm	2.5' × 2.5'	Monthly	World-Clim
Potential Evaporation	m	0.1º × 0.1º	Monthly	ERA5
Shortwave radiation	J/m^2	0.1º × 0.1º	Monthly	ERA5
Surface latent heat flux	J/m^2	0.1º × 0.1º	Monthly	ERA5
Surface sensible heat flux	J/m^2	0.1º × 0.1º	Monthly	ERA5
Dewpoint temperature	ºC	0.1º × 0.1º	Monthly	ERA5
Relative humidity	%	0.25º × 0.25º	Monthly	ERA5
Net primary Productivity	kgC/m^2	500 m × 500 m	Annual	MODIS
Digital Elevation Model	m	30 m × 30 m	-	EOS-DIS
Land Use & Land Cover	-	1 km × 1 km	Multi-Annual	SIOSE

Table 2 Models hyperparameter tuning for cross-validation

Model	Hyperparameter	Range	By
RF	Number of variables to split at node	[2, 58]	1
	Minimum node size	1, 3, 5, 10	-
GBT	Number of trees	[50, 500]	50
	Maximal depth	0, [3, 8]	1
	Learning rate	0.01, 0.1	-
MLPNN	Number of hidden layers	[1, 9]	2
	Number of ensemble members	√5301	-

For each year of the study period, data was extracted from a total of 400 random sampling points. The 70% of this was used to train and validate the models through cross-validation. The training data was split in three and combined in proportion of 2:1 for training and validating the models hyperparameters (Table 2) a total of three folds. Each split was selected as validation dataset on each fold, while the 2 remaining were joint for training. We used the lowest Root Mean Squared Error (RMSE) (Eq. 1) criteria to assess the performance of the models, but the Mean Absolute Error (MAE) (Eq. 2) and the squared correlation (R^2) (Eq. 3) were also provided.

$$RMSE = \sqrt{1/n \sum_{i=1}^{n} (y_i - \widehat{y_i})} \quad (1)$$

Table 3 Characterisation of the proposed scenarios for the forecast of the Carbon absorption

Scenario	Climatic projection	Land uses
Scenario 1	Historical	Forest
Scenario 2	Historical	Agriculture
Scenario 3	SSP2-4.5	Unchanged
Scenario 4	SSP2-4.5	Forest
Scenario 5	SSP2-4.5	Agriculture
Scenario 6	SSP5-8.5	Unchanged
Scenario 7	SSP5-8.5	Forest
Scenario 8	SSP5-8.5	Agriculture

$$MAE = 1/n \sum_{i=1}^{n} |y_i - \widehat{y_i}| \quad (2)$$

$$R^2 = 1 - \frac{\sum (y_i - \widehat{y_i})^2}{\left(y_i - \bar{y}\right)^2} \quad (3)$$

where n is the number of observations; y_i and $\hat{y}_i$ are respectiveley the observed and predicted values of the i observation; and $\bar{y}$ is the mean of y.

After the cross-validation, the models were tested using the remaining 30% of the sampled dataset, according to the lowest RMSE – highest R^2 criteria. Each model was applied to the whole testing dataset as well as to each area individually. To assess the transferability of the model, the models were applied to a dataset consisting of a sample of 400 points per year in the whole Galician territory. This data was filtered by erasing all the sampling points which did not fall in an area whose land use was in the training dataset.

2.4 Carbon Absorption Forecast Under Future Scenarios

A total of eight scenarios (Table 3) were designed combining different land uses with climatic projections for the period ranging from 2026 to 2050. The first and second scenarios consisted in keeping the historic records of climatic variables as hindcast and applying land use change as only driving factor affecting the carbon absorption. The proposed land use scenarios were labelled as "forest" and "agricultural". Each one of them consisted of setting the 70% of any land uses that did not fall into the "target" category as thus. For example, for the "forest" scenario, we set 70% of data which was not categorised as forest to such category. Same with agriculture.

On their part, third, fourth and fifth scenarios were set under the Shared Socioeconomical Pathway SSP2-4.5, by keeping the land use as constant from the last sampling instant (2014), and by applying the same changes to land use as in scenarios 1 and 2. Finally, the scenarios 6, 7 and 8

were set as the third, fourth and fifth, but under the SSP5-8.5, representing the fossil fuel-based development.

Data on monthly mean maximum and mean minimum temperature, accumulated precipitation, shortwave downwelling radiation, latent and sensible heat fluxes and relative humidity were retrieved from the Coupled Model Intercomparison Project (CMIP6) catalogue (Service 2021), regarding the representative pathways of SSP2-4.5 and SSP5-8.5. The dewpoint temperature was deterministically obtained as the result of Eq. 4.

$$T_d = T - \frac{100 - U}{5} \tag{4}$$

where T_d is the dewpoint and T is the actual temperature, and U is the relative humidity.

After applying the models to data from scenarios, an ANOVA test between the results obtained for each scenario was carried to evaluate the influence of them on the overall carbon absorption of the ecosystem. In the cases the ANOVA confirmed that there is a true significative difference in the NPP means, a Tukey test was performed to evaluate which scenarios held the significantly different values. In addition, the trends and their significance were assessed through linear regression modelling.

3 Results and Discussion

3.1 Models Performance

The models generally performed with low errors and high correlation between the predictions and the observed data. Thus, the RMSE was generally around the 5% of the mean value of the NPP, while the correlations were between 0.8 and 0.9 (Table 4). During the cross-validation stage, the GBT outperformed the other algorithms, obtaining lower RMSE and Mean Absolute Error (MAE); and higher correlation (Table 4). The most optimal hyperparameters for this model were 500 trees with maximal depth of 5, and a learning rate of 0.1. Compared to Bai et al. (2021), our results show a slightly lower correlation between actual and modelled data in the validation stage. They tried two gradient boosting methods and obtained squared correlations of 0.98 for the GBT and 0.96 for the LightGBM, facing our 0.90. This implies that the suppression of the Remote-Sensing variables in the training dataset may lead to worse predictions. However, it makes the model applicable to future climatic scenarios, which was the main scope of this work.

Table 4 Performance of the models in the training stage, with 3-folded cross-validation

Parameter	RF	GBT	MLPNN
RMSE (kgC/m^2)	0.07 ± 0.00	0.05 ± 0.00	0.06 ± 0.00
MAE (kgC/m^2)	0.05 ± 0.00	0.04 ± 0.00	0.04 ± 0.00
R^2	0.80 ± 0.01	0.90 ± 0.00	0.86 ± 0.00

On its part, the most optimal tuning for the RF was 55 variables to possibly split at each node with a minimum node size of 5. This combination of hyperparameters resulted into a RMSE of 0.07 ± 0.00 and R^2 of 0.80 ± 0.01. These results were in line with Wei et al. (2017), which obtained a R^2 of 0.79 through a similar approach; although their combination of RF with LUE models resulted in worst predictions ($R^2 = 0.67$). Nonetheless, the results by Bai et al. (2021), outperformed, once again, the ones we obtained, as their RF produced predictions that explained the 95% of the observations. On the other hand, the MLPNN optimal tuning were 7 hidden layers with the square root of the number of samples ($n = 5301$). The RMSE in this case was 0.06 ± 0.00 and the R^2 was 0.86 ± 0.00. This result was in line with Melesse and Hanley (2005), who obtained correlations of 0.89 and 0.83 in the training stage of a similar model, respectively in forest and grassland areas. Their model, however, outperformed our results in wheat areas, obtaining a R^2 of 0.89. It is worth mentioning that the computational cost of training the RF was lower than for the other two methods. In fact, the MLPNN had a remarkable high computational demand due to the large number of ensemble members (72).

The models' performance was tested separately in all the sampling sites, as well as in the combination of all of them. Also, a random sampling along all the Autonomous Community of Galicia was set as blank to assess the transferability of the models (Table 5). Generally, the GBT algorithm outperformed the others in the testing stage, both,

Table 5 Performance of the models in the testing stage in different sampling sites

Testing site	RF		GBT		MLPNN	
	RMSE (kgC/m^2)	R^2 -	RMSE (kgC/m^2)	R^2 -	RMSE (kgC/m^2)	R^2 -
All	0.06	0.88	0.04	0.95	0.08	0.77
Central Massif	0.05	0.78	0.03	0.91	0.07	0.62
Fragas do Eume	0.04	0.83	0.03	0.87	0.06	0.48
Sil Canyon	0.04	0.79	0.03	0.90	0.05	0.67
Tambre River	0.06	0.66	0.04	0.86	0.08	0.46
Transferability	0.09	0.71	0.09	0.71	0.11	0.59

separately on each study area and in the ensemble of all of them. For this model, the predictions explained more than 80% of real data in all the testing datasets, and a remarkable above 90% in the Central Massif and the whole testing set (Table 5). Compared to the others, the GBT performed with outstandingly low error and high correlation in all sites, but especially in the Tambre River testing data. Therefore, we applied the GBT model for the forecast of NPP in all the areas in the future scenarios.

The performance of the models was worsened when applied to a set of sampling points in the whole Galician territory. This is assumed to be consequence of the lack of training samples of land uses apart of forestry, agriculture, and "not in other economic use". Moreover, the latter is, perhaps, a too generic use to properly train the models. The Root Mean Squared Errors reached more than double their values in the training areas, and the R^2 were reduced from around 0.9 to 0.71 or even less in the case of the MLPNN. Thus, the transferability of the models to other areas was deprecated, despite their generally high performance within the limits of the study areas.

3.2 NPP Forecast Under Future Scenarios

Based on the GBT model, a total of 8 scenarios were set, and an ANOVA test was carried to assess the significance of the results (Table 6). These results showed that there is generally a significant difference in the average carbon absorption depending on the land use and climatic pathway. The only non-significant difference in the means was obtained when comparing the NPP of the proposed land uses for the SPP5-8.5 (Table 6). For this climatic pathway, and depending on the land use, a decrease of the NPP between the 5 and the 5.5% is expected, compared to the historical average. On the other hand, the differences in the NPP mean between land use were significant for the SPP2-4.5 pathway as well as in the hypothetical changes in the hindcast data. In both cases, the difference in the carbon absorption is between the unchanged land use and the proposed resulted significative with p-values lower than 0.01. Nonetheless, the differences between the two proposed land use changes are non-significant both, in the hindcast and the SSP2-4.5 data. In any case, the carbon absorption for the SSP2-4.5 pathway is expected to decrease around a 6.9% from the historical mean (Table 6).

Table 6 Mean ± Sd. Of the predicted net primary Productivity (kgC/m^2) for each scenario

LULC	Climatic pathways			
	Historical	SSP2-4.5	SSP5-8.5	*p*-value
Unchanged	0.88 ± 0.16	0.82 ± 0.17	0.83 ± 0.17	***
Reforestation	0.89 ± 0.15	0.84 ± 0.16	0.84 ± 0.17	***
Agricultural	0.89 ± 0.15	0.83 ± 0.15	0.84 ± 0.17	***
p-value	***	***	0.164	

*** p-value < 0.001

** 0.001 ≤ p-value < 0.01

* 0.01 ≤ p-value < 0.05

. 0.05 ≤ p-value < 0.1

Regarding land use, the main class SIOSE indexing in the sampling areas, namely, "areas not in any other economic use", was too generic to properly assess the influence of land use on the NPP. Furthermore, the only other two classes with samples enough to be significant were "agricultural" and "forest", which according to our results, do not have such a big difference in terms of NPP. This is, most probably, a result of the fact that the carbon absorption depends on the plant species as some authors explain (Heijmans et al. 2004; Hernandez-Ramirez et al. 2011), but our model was not trained to distinguish between them. In addition, the information of Land Use was kept constant during the years when there were no records (i.e. from 2005 to 2009 we considered land use to be as in 2005), which may not stick to reality in every case. Moreover, a probably determining fact that may have caused such counter-intuitive results is the land use classification itself. We selected the Hierarchical INSPIRE Land Use Classification System (HILUCS) of the SIOSE for its simplicity, distinguishing between only 16 classes which we considered convenient for our research. However, as stated above, the land use with highest occurrence rate in the pilot areas was categorised as "areas not in any other economic use" which is not defining any specific characteristic that could affect the values of carbon absorption of the ecosystem. Furthermore, landscape in the Autonomous Community of Galicia is generally highly fragmented and this information may have been lost in the resampling process. These might have led to an underestimation of the land use importance by the model. In this sense, a refinement of the model should consist of testing some other available land use or land cover classifications. Some other classifications that could be interesting for that purpose are other classifications provided by SIOSE, the Corine Land Cover or the MODIS Land Cover type, among others.

No significant linear trends were observed in the historical data nor any of the SSP2-4.5 projections. However, for all the SSP5-8.5 projections, the reduction on the NPP shows a significant trend of -0.003** kgC/m^2 per year for any land use (Fig. 3). This implies that the ability for Carbon Absorption of Galician interior forests is not only expected to get reduced towards the near future under the SSP2-4.5 and the SSP5-8.5, but also, for the latter, it is forecasted to stagnate even more in the mid and long-term future. This is, in fact, opposite to what would be expected from the definition of the modelled SSPs, as the former represents a future where there is no change in the Carbon emissions until the

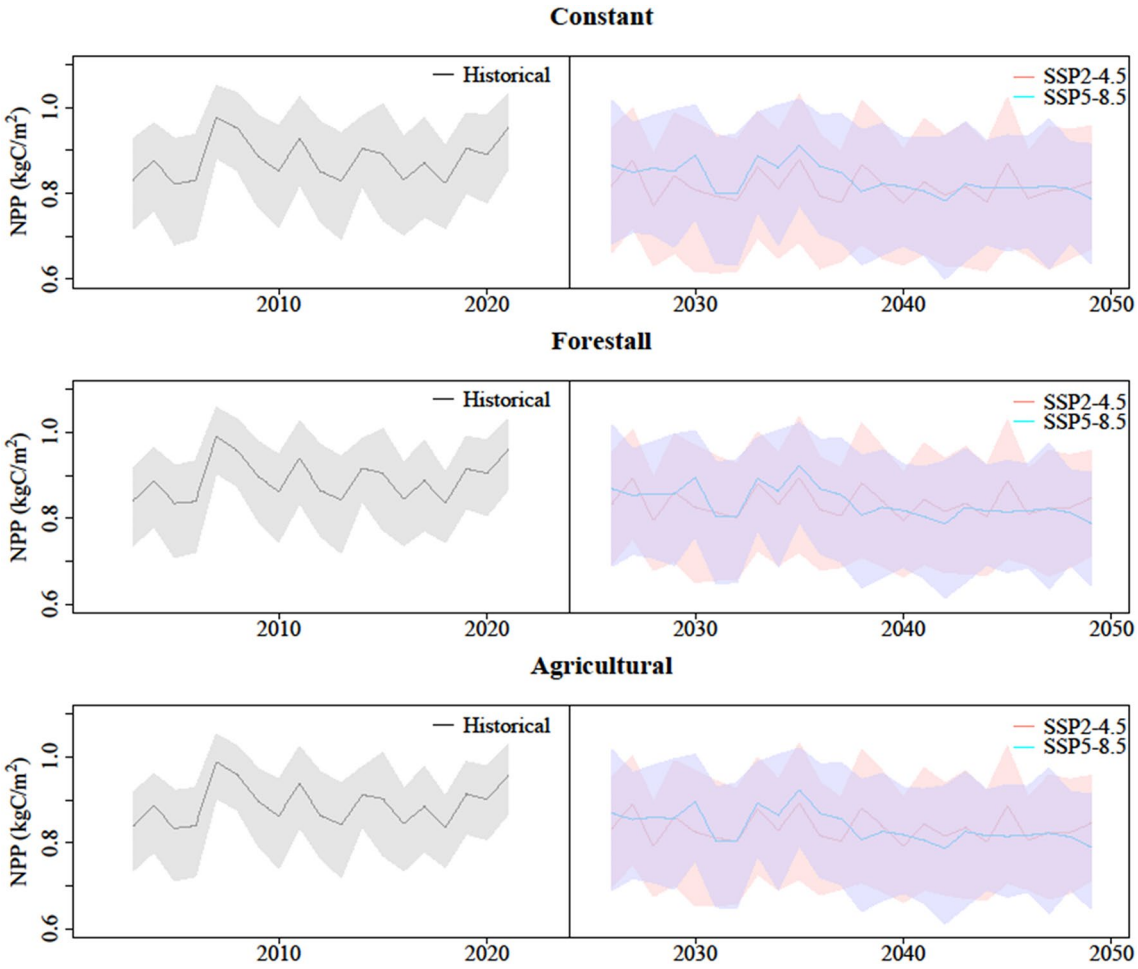


Fig. 3 Forecasted time series of NPP for each scenario

Table 7 Mean ± Sd. Of the predicted net primary Productivity (kgC/m^2) on each sampling area

Climate	Land use	Central Massif	Sil Canyons	Fragas do Eume	Tambre River
Historical	Unchanged	0.73 ± 0.11	0.79 ± 0.07	1.12 ± 0.09	0.95 ± 0.11
Historical	Forest	0.76 ± 0.10	0.80 ± 0.06	1.12 ± 0.08	0.96 ± 0.09
Historical	Agriculture	0.76 ± 0.10	0.80 ± 0.06	1.13 ± 0.08	0.95 ± 0.09
SSP2-4.5	Unchanged	0.63 ± 0.05	0.63 ± 0.03	0.90 ± 0.10	0.97 ± 0.06
SSP2-4.5	Forest	0.67 ± 0.05	0.66 ± 0.03	0.91 ± 0.09	0.98 ± 0.06
SSP2-4.5	Agriculture	0.66 ± 0.05	0.65 ± 0.03	0.91 ± 0.09	0.97 ± 0.06
SSP5-8.5	Unchanged	0.66 ± 0.06	0.69 ± 0.04	0.96 ± 0.13	0.97 ± 0.08
SSP5-8.5	Forest	0.67 ± 0.06	0.70 ± 0.04	0.96 ± 0.13	0.97 ± 0.08
SSP5-8.5	Agriculture	0.67 ± 0.06	0.70 ± 0.04	0.96 ± 0.13	0.97 ± 0.09

year 2050, and the latter is a fossil fuel-based scenario in which the emissions of GHG significantly rise (Meinshausen et al. 2020), increasing not only the temperature but also the atmospheric Carbon availability.

The model was used to estimate the NPP of all the sampling areas separately and under the same scenarios (Table 7). The historical data showed Fragas do Eume to be the area with the highest Carbon storage capacity, followed by the Tambre River, then the Sil Canyons and finally the Central Massif. Although the majority of these areas are expected to record a decrease in the mean NPP for the SSP2-4.5 and the SSP5-8.5, Fragas do Eume and the Sil Canyons are the areas where the Carbon absorption is predicted to deteriorate the most. The former is expected to record a decrease of a 20% of its current capacity for the SSP2-4.5 and 14% for the SSP5-8.5. This area is the only registering a significant linear trend, which gets a value of -0.008 kgC/m^2 per year for the SSP5-8.5. The Sil Canyons, on their part, are predicted to reduce their average NPP in a 20% and in a 13% respectively for the SSP2-4.5 and the SSP5-8.5,

but not significant trend was observed. On the other hand, a decrease on the average NPP of a respective 14% and 10% is expected for the Central Massif. In contrast, the only area which is projected to record a slight increment of the Carbon storage is the Tambre River. This increment, however, is lesser than the decrease in the other areas and its magnitude does not depend on the climatic pathway. Generally, land uses do not seem to have an actual influence on the NPP; nonetheless, for the SSP2-4.5, in the Sil Canyons and the Central Massif, there is a difference between keeping the land use constant and setting most of the area either as forest or agricultural.

Given the definition of NPP, which is the energy generated through photosynthesis minus the plant respiration, and in the insights that the photosynthesis rate is generally inclined to rise with temperature; the results here obtained imply that the plant respiration may be rising at a higher rate with climate change, at least in the areas where NPP is forecasted to decline in the future. This opens the gate to a potential research line to assess the extent to which plant respiration influences the Ecosystem Primary Productivity trend related to climatic variables.

3.3 General Discussion

We trained a Machine Learning model for the forecasting of Net Primary Productivity as a quantification of the carbon absorption. Our best fitting model, the GBT, resulted in a high accuracy, with an RMSE of 0.05 kgC/m^2 and correlation of 0.90 in the cross-validation and RMSE of 0.04 kgC/m^2 and R^2 of 0.95 in the testing stage, which outperformed the RF (RMSE = 0.07, R^2 = 0.80) and the MLPNN (RMSE = 0.06, R^2 = 0.86) both in the training and the testing stages. The models' performance was worsened when we tried to transfer them to other areas, especially for the MLPNN, which also had the issue of the high computing demand. Compared to similar studies trained with data from remote sensing sources, the GBT here developed generally outperformed other models in terms of correlation, such as ANNs (Papale and Valentini 2003), SVM (Yang et al. 2007), RF (Wei et al. 2017) or Cubist (Xiao et al. 2008). On the other hand, Piecewise regression (PWR) (Zhang et al. 2007), Gaussian Process Regression (GPR) (Reyes-Muñoz et al. 2024), Support Vector Regressor (SVR) (Ueyama et al. 2013), Model Tree Ensembles (MTE) (Jung et al. 2011; Liang et al. 2017) or other GBT methods (Bai et al. 2021) obtained predictions that explained more than 90% of the real data, similar or even better than our results. These, however, were trained using Remote Sensing data, making them unavailable for future scenario forecasting. It must be noted that the target of the studies here compared is to predict the carbon absorption, but it is quantified differently for each one of them (i.e. GPP, NEE, NEP, NPP, etc.). Therefore, we used R^2 as a normalised, dimensionless comparative parameter, as the RMSE depends on the variable units.

Our model showed a potential decrease in the carbon absorption in the short-term future (2026–2050), no matter the predominant land use or the climatic pathway. There is, however, a significant difference between the means of absorbed CO_2 of each scenario. The model estimates the reduction of NPP to have a lesser extent for the SSP5-8.5 than for the SSP2-4.5. Despite this may seem counterintuitive, as a greater radiative forcing should imply a greater shock to the ecosystem, the temperature seasonality and range seem to be related with this, as for example the temperature seasonality not only exhibits a negative trend with the NPP, but also is projected to record wider ranges on SSP5-8.5 than on SSP2-4.5, which may be influencing the results. Another hypothesis is that the definition of NPP itself is related to this, as it does not account the energy due to respiration, differently as GPP, which other authors have predicted to increase. These, however, are just hypotheses, as the coupling of the components of the climate system is more complex and need further study, which is out of scope for this research. In this sense, an approach based on numeric models is proposed for further research.

Nevertheless, there exists a significant trend to reduction in the long-term future for the former, which does not seem to exist in the latter. This regional differences in the carbon absorption contradicts the global historical trend, which is to rise with temperature (Chen et al. 2024), CO_2 concentration (Schlund et al. 2020), and the global projected trends of the Net Ecosystem Exchange, which favours the CO_2 uptake by the terrestrial biosphere (Zhang et al. 2023). Moreover, similar studies in other areas forecasted an increase in the carbon absorption for the future, rising with the radiative forcing. Wang et al. (2024) predicted an augmentation of the GPP of 10.3–19.5% for the SSP2-4.5 and of 19.7–46.8% for the SSP5-8.5 in the state of NSW, in Australia. These, notwithstanding, do not necessarily fall into contradiction with our findings. First, the time series of a specific variable in a few small regions in the Northwestern Spain are not expected to explain the overall planetary carbon absorption, especially when this, as a concept, may be quantified by different variables (i.e. NPP, GPP, NEE, etc.). On the other hand, in this study we mostly used agrometeorological variables to train the models, while omitted some other potentially interesting variables, as could be the atmospheric carbon concentration, or soil related data. Thus, this work is representative of how these agrometeorological variables affect the NPP, independently from other factors.

The model here developed also predicted that for any pathway, allocating more land area to forest or agricultural does not make a significant difference in the carbon

absorption. This, nonetheless, does not mean that agriculture and forest areas have the same impact on the ecosystem and environmental resources. Intuitively, agriculture and related land uses entail greater environmental costs than forestry (Tilman 1999); however, our model was not able to record these differences in terms of carbon absorption, probably due to the lack of vegetation type data for each land use class. This means, that the SIOSE classification of land use (HILUCS) may be too generic for this kind of work and so, further research on the topic is suggested.

Furthermore, this study was carried making use of reanalysis data, which must be taken with caution in some cases (Parker 2016), specifically in issues related to moisture budget (Nigam and Ruiz-Barradas 2006). Besides, we used potential evaporation data obtained from the ERA5 reanalysis for training, which in first place is calculated setting the vegetation parameters as "crops and mixed farming" and presuming "no stress from soil moisture" (Hersbach et al. 2018), which may have interfered in the results of the model regarding the land use changes. On the other hand, for the application of the scenarios we used evaporation data from the CMIP6 and considered this to be equivalent to the former. Moreover, this data was of too low spatial resolution in comparison to the resolution of the target variable. Data with too low resolution, as the data provided by the ERA5 was resampled to a higher resolution through the nearest neighbour technique. When the difference in the target and the input resolution was smaller, we used the bilinear interpolation instead. However, due to the high computational demand of such process, it was not applied to create the dataset for the second testing stage, we simply extracted the values from the original layers, which should not imply that much of a difference as the interpolation method was the nearest neighbour technique.

Despite all the above-mentioned issues, the methodology here applied was proven to be a powerful tool for the prediction and forecast of the carbon absorption in terms of NPP. Thereby, its application is recommended for the management of forest areas and further study is advised, especially in terms of land use, with perhaps a more specific classification, or a more heterogeneous sample for training. In this issue, the combination of this methodology with the InVEST software is regarded as an interesting matter for future studies, as getting to know how land use will affect the results could imply a more sustainable management of forest areas. In this sense, the methodology here developed, with a little bit of refinement, stands as a useful tool for policy makers and administrations to assess the impacts of certain activities on the overall ecosystem carbon absorption, so as the usage of such areas could be carried in a more responsible way, as they play a key role as sinks in the carbon cycle.

4 Conclusions

In this study, we trained a set of different Machine Learning algorithms to predict the combined NPP of four differentiated Protected Areas of the Autonomous Community of Galicia. The most optimal for the prediction of the carbon absorption was the Gradient Boosting Machine. This model was then tested in the areas where the training data was obtained from, and it reached a high performance. The model was tested on data from the areas where it was trained, providing accurate results. Besides, a second testing stage was carried to assess the model's transferability, picking random points all around the region of Galicia; however, the performance of the model was affected negatively in this stage. It was implemented to several future scenarios, and the results showed a significant difference through the different climatic pathways, expecting a decrease of the NPP for all of them. On the other hand, the differences on the means of the land use changes were only significant for the historical period and the SSP2-4.5, and in both cases there was not a significant difference between the forest and the agricultural-focused land use changes. Lastly, the NPP time series showed a significative decreasing linear trend for the SPP5-8.5 forecast, while no significant trends were observed in the other climatic scenarios. This methodology is a powerful tool for providing technical advice for the management of forest areas; however, combining it with other methods and software, could imply an improvement on its reliability.

Funding This research is part of the Grant PID2022-138374OA-I00 funded by MICIU/AEI/https://doi.org/10.13039/501100011033/ and ERDF, EU (P. Lago); by Grant TED2021-130241 A-I00 funded by MCIN/AEI/ https://doi.org/10.13039/501100011033 and by the "European Union NextGenerationEU/PRT" (X. Álvarez).; and partially funded by the Postdoctoral Program of Xunta de Galicia, ED481B-2023-042 (C. Acuña-Alonso). Funding for open access charge was from the Universidade de Vigo/CISUG.

Author Contributions Pedro Lago: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing-Original Draft, Visualization. Carolina Acuña: Conceptualization, Investigation, Writing-Original Draft, Writing-Review & Editing, Visualization, Supervision. Xana Álvarez: Conceptualization, Investigation, Writing-Review & Editing, Visualization, Supervision, Project administration, Funding acquisition.

Funding Open Access funding provided thanks to the CRUE-CSIC agreement with Springer Nature.

Data Availability Data derived from this research, along with a description, can be found at the following link https://github.com/plaglez/NPP-scenario-forecast_PAs-gal/tree/data.

Declarations

Conflict of Interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

Almeida B, David J, Campos FS, Cabral P (2024) Satellite-based Machine Learning modelling of Ecosystem Services indicators: a review and meta-analysis. Appl Geogr 165(March). https://doi.org/10.1016/j.apgeog.2024.103249

AppEEARS Team (2024) *Application for Extracting and Exploring Analysis Ready Samples (AppEEARS). Ver. 3.49. NASA EOSDIS Land Processes Distributed Active Archive Center (LP DAAC)*. https://appeears.earthdatacloud.nasa.gov

Bai Y, Liang S, Yuan W (2021) Estimating global gross primary production from sun-induced chlorophyll fluorescence data and auxiliary information using machine learning methods. Remote Sens 13(5):1–22. https://doi.org/10.3390/rs13050963

Breiman L (2001) Random forests. Mach Learn 45(1):5–32. https://doi.org/10.1023/A:1010933404324

Brown TC, Bergstrom JC, Loomis JB (2007) Defining, valuing, and providing ecosystem goods and services. Nat Resour J 47(2):329–376

Cary GJ, Keane RE, Gardner RH, Lavorel S, Flannigan MD, Davies ID, Li C, Lenihan JM, Rupp TS, Mouillot F (2006) Comparison of the sensitivity of landscape-fire-succession models to variation in terrain, fuel pattern, climate and weather. Landscape Ecol 21(1):121–137. https://doi.org/10.1007/s10980-005-7302-9

Chas-Amil ML, Touza J, García-Martínez E (2013) Forest fires in the wildland-urban interface: a spatial analysis of forest fragmentation and human impacts. Appl Geogr 43:127–137. https://doi.org/10.1016/j.apgeog.2013.06.010

Chen B, Zou H, Zhang B, Zhang X, Wang C, Zhang X (2023) Distribution change and protected area planning of Tilia amurensis in China: a study of integrating the climate change and present habitat landscape pattern. Global Ecol Conserv 43:e02438. https://doi.org/10.1016/j.gecco.2023.e02438

Chen Y, Zhu Z, Zhao W, Li M, Cao S, Zheng Y, Tian F, Myneni RB (2024) The direct and indirect effects of the environmental factors on global terrestrial gross primary productivity over the past four decades. Environ Res Lett 19(1). https://doi.org/10.1088/1748-9326/ad107f

Consellería de Medio Ambiente Territorio e Infraestructuras (2014) *DECRETO 37/2014, do 27 de marzo, polo que se declaran zonas especiais de conservación os lugares de importancia comunitaria de Galicia e se aproba o Plan director da Rede Natura 2000 de Galicia*

Daily GC (1997) Introduction: What Are Ecosystem Services? *NATURE'S SERVICES: SOCIETAL DEPENDENCE ON NATURAL EcOSYsTEMs*, *1*(3)

Duncanson L, Liang M, Leitold V, Armston J, Krishna Moorthy SM, Dubayah R, Costedoat S, Enquist BJ, Fatoyinbo L, Goetz SJ, Gonzalez-Roglich M, Merow C, Roehrdanz PR, Tabor K, Zvoleff A (2023) The effectiveness of global protected areas for climate change mitigation. Nat Commun 14(1). https://doi.org/10.1038/s41467-023-38073-9

Easterling DR, Meehl GA, Parmesan C, Changnon SA, Karl TR, Mearns LO (2000) *Climate Extremes: Observations, Modeling, and Impacts*. *289*(September), 2068–2074

Estrada GCD, Soares MLG (2017) Global patterns of aboveground carbon stock and sequestration in mangroves. Anais Da Acad Brasileira De Ciencias 89(2):973–989. https://doi.org/10.1590/0001-3765201720160357

EU. (2004) *2004/813/EC: Commission Decision of 7 December 2004 adopting, pursuant to Council Directive 92/43/EEC, the list of sites of Community importance for the Atlantic biogeographical region (notified under document number C(2004) 4032)*

Fahrig L (2003) Effects of Habitat Fragmentation on Biodiversity. Annu Rev Ecol Evol Syst 34:487–515. https://doi.org/10.1146/annurev.ecolsys.34.011802.132419

Fick SE, Hijmans RJ (2017) WorldClim 2: new 1-km spatial resolution climate surfaces for global land areas. Int J Climatol 37(12):4302–4315. https://doi.org/10.1002/JOC.5086

Flannigan MD, Wotton BM, Marshall GA, de Groot WJ, Johnston J, Jurko N, Cantin AS (2016) Fuel moisture sensitivity to temperature and precipitation: climate change implications. Clim Change 134(1–2):59–71. https://doi.org/10.1007/s10584-015-1521-0

Friedman JH (2001) Greedy function approximation: a gradient boosting machine. Ann Stat 29(5):1189–1232. https://doi.org/10.1214/aos/1013203451

Gauquelin T, Michon G, Joffre R, Duponnois R, Génin D, Fady B, Bou Dagher-Kharrat M, Derridj A, Slimani S, Badri W, Alifriqui M, Auclair L, Simenel R, Aderghal M, Baudoin E, Galiana A, Prin Y, Sanguin H, Fernandez C, Baldy V (2018) Mediterranean forests, land use and climate change: a social-ecological perspective. Reg Envriron Chang 18(3):623–636. https://doi.org/10.1007/s10113-016-0994-3

Gautam S, Mishra U, Scown CD, Wills SA, Adhikari K, Drewniak BA (2022) Continental United States may lose 1.8 petagrams of soil organic carbon under climate change by 2100. Glob Ecol Biogeogr 31(6):1147–1160. https://doi.org/10.1111/geb.13489

Harris NL, Gibbs DA, Baccini A, Birdsey RA, de Bruin S, Farina M, Fatoyinbo L, Hansen MC, Herold M, Houghton RA, Potapov PV, Suarez DR, Roman-Cuesta RM, Saatchi SS, Slay CM, Turubanova SA, Tyukavina A (2021) Global maps of twenty-first century forest carbon fluxes. Nat Clim Change 11(3):234–240. https://doi.org/10.1038/s41558-020-00976-6

Heijmans MMPD, Arp WJ, Chapin FS (2004) Carbon dioxide and water vapour exchange from understory species in boreal forest. Agric for Meteorol 123(3–4):135–147. https://doi.org/10.1016/j.agrformet.2003.12.006

Hernandez-Ramirez G, Hatfield JL, Parkin TB, Sauer TJ, Prueger JH (2011) Carbon dioxide fluxes in corn-soybean rotation in the midwestern U.S.: Inter- and intra-annual variations, and biophysical controls. *Agricultural and Forest Meteorology*, *151*(12), 1831–1842. https://doi.org/10.1016/j.agrformet.2011.07.017

Hersbach H, Muñoz Sabater J, Nicolas, Rozum I, Simmons, Vamborg F, Bell A, Berrisford B, Biavati P, Buontempo G, Horányi C, Peubey AJ, Radu C, Schepers R, Soci D, Dee C, D., Thépaut J-N (2018) *Essential climate variables for assessment of climate variability from 1979 to present. Copernicus Climate Change Service (C3S) Data Store (CDS).*

Hijmans RJ, Phillips S, Leathwick J, Elit J (2023) *Package dismo*.https://rspatial.org/raster/sdm/

Instituto Geográfico Nacional (2014) *Sistema de Ocupación del Suelo de España*. http://centrodedescargas.cnig.es/CentroDescargas/catalogo.do?Serie=SIOSE

Johnson MP (2016) Photosynthesis. Essays Biochem 60(3):255–273. https://doi.org/10.1042/EBC20160016

Joppa LN, Loarie SR, Pimm SL (2008) On the protection of protected areas. Proc Natl Acad Sci USA 105(18):6673–6678. https://doi.org/10.1073/pnas.0802471105

JPL NASA (2013) NASA Shuttle Radar Topography Mission Global 3 arc second NetCDF. https://doi.org/10.5067/MEaSUREs/SRTM/SRTMGL3_NC.003

Jung M, Reichstein M, Margolis HA, Cescatti A, Richardson AD, Arain MA, Arneth A, Bernhofer C, Bonal D, Chen J, Gianelle D, Gobron N, Kiely G, Kutsch W, Lasslop G, Law BE, Lindroth A, Merbold L, Montagnani L, Williams C (2011) Global patterns of land-atmosphere fluxes of carbon dioxide, latent heat, and sensible heat derived from eddy covariance, satellite, and meteorological observations. J Geophys Research: Biogeosciences 116(3):1–16. https://doi.org/10.1029/2010JG001566

Lele S, Wilshusen P, Brockington D, Seidler R, Bawa K (2010) Beyond exclusion: alternative approaches to biodiversity conservation in the developing tropics. Curr Opin Environ Sustain 2(1–2):94–100. https://doi.org/10.1016/j.cosust.2010.03.006

Liang W, Lü Y, Zhang W, Li S, Jin Z, Ciais P, Fu B, Wang S, Yan J, Li J, Su H (2017) Grassland gross carbon dioxide uptake based on an improved model tree ensemble approach considering human interventions: global estimation and covariation with climate. Glob Change Biol 23(7):2720–2742. https://doi.org/10.1111/gcb.13592

Ma Z, Chen W, Xiao A, Zhang R (2023) The susceptibility of Wetland areas in the Yangtze River Basin to temperature and vegetation changes. Remote Sens 15(18). https://doi.org/10.3390/rs15184534

Meinshausen M, Nicholls ZRJ, Lewis J, Gidden MJ, Vogel E, Freund M, Beyerle U, Gessner C, Nauels A, Bauer N, Canadell JG, Daniel JS, John A, Krummel PB, Luderer G, Meinshausen N, Montzka SA, Rayner PJ, Reimann S, Wang RHJ (2020) The shared socioeconomic pathway (SSP) greenhouse gas concentrations and their extensions to 2500. Geosci Model Dev 13(8):3571–3605. https://doi.org/10.5194/gmd-13-3571-2020

Melesse AM, Hanley RS (2005) Artificial neural network application for multi-ecosystem carbon flux simulation. Ecol Model 189(3–4):305–314. https://doi.org/10.1016/j.ecolmodel.2005.03.014

MITECO (2021) *Anuario de Estadística Forestal* (p. 79)

Muñoz Sabater J (2019) ERA5-Land monthly averaged data from 1950 to present. Copernicus Clim Change Service (C3S) Clim Data Store (CDS). https://doi.org/10.24381/cds.68d2bb30

Nigam S, Ruiz-Barradas A (2006) Seasonal hydroclimate variability over North America in global and regional reanalyses and AMIP simulations: varied representation. J Clim 19(5):815–837. https://doi.org/10.1175/JCLI3635.1

Papale D, Valentini R (2003) A new assessment of European forests carbon exchanges by eddy fluxes and artificial neural network spatialization. Glob Change Biol 9(4):525–535. https://doi.org/10.1046/j.1365-2486.2003.00609.x

Parker WS (2016) Reanalyses and observations: what's the difference? Bull Am Meteorol Soc 97(9):1565–1572. https://doi.org/10.1175/BAMS-D-14-00226.1

Pham HC, Alila Y (2024) Science of forests and floods: the quantum leap forward needed, literally and metaphorically. Sci Total Environ 912:169646. https://doi.org/10.1016/j.scitotenv.2023.169646

Reyes-Muñoz P, Kovács D, Berger D, Pipia K, Belda L, Rivera-Caicedo S, J. P., Verrelst J (2024) Inferring global terrestrial carbon fluxes from the synergy of Sentinel 3 & 5P with gaussian process hybrid models. Remote Sens Environ 305(February):114072. https://doi.org/10.1016/j.rse.2024.114072

Rodrigues ASL, Akçakaya HR, Andelman SJ, Bakarr MI, Boitani L, Brooks TM, Chanson JS, Fishpool LDC, Fonseca D, Gaston GAB, Hoffmann KJ, Marquet M, Pilgrim PA, Pressey JD, Schipper RL, Sechrest J, Stuart W, Underhill SN, Waller LG, Yan RW, X (2004) Global gap analysis: Priority regions for expanding the global protected-area network. Bioscience 54(12):1092–1100. https://doi.org/10.1641/0006-3568(2004)054[1092:GGAPRF]2.0.CO;2

Rumman R, Atkin OK, Bloomfield KJ, Eamus D (2018) Variation in bulk-leaf 13 C discrimination, leaf traits and water-use efficiency–trait relationships along a continental-scale climate gradient in Australia. Glob Change Biol 24(3):1186–1200. https://doi.org/10.1111/gcb.13911

Running S, Zhao M (2021) MODIS/Terra Net Primary Production gap-filled yearly L4 global 500m SIN Grid V061. NASA EOSDIS Land Processes Distrib Act Archive Cent. https://doi.org/10.5067/MODIS/MOD17A3HGF.061

Schlund M, Eyring V, Camps-Valls G, Friedlingstein P, Gentine P, Reichstein M (2020) Constraining uncertainty in projected gross primary production with machine learning. J Geophys Research: Biogeosciences 125(11). https://doi.org/10.1029/2019JG005619

Service CCC (2021) C. D. S. *CMIP6 climate projections. Copernicus Climate Change Service (C3S) Climate Data Store (CDS)*. https://doi.org/10.24381/cds.c866074c

Shi Z, Ferrari G, Ai P, Marinello F, Pezzuolo A (2024) Bioenergy potential from agricultural by-product in 2030: an AI-based spatial analysis and climate change scenarios in a Chinese region. J Clean Prod 436(December 2023):140621. https://doi.org/10.1016/j.jclepro.2024.140621

Taylor FW (2002) The greenhouse effect and climate change revisited. Rep Prog Phys 65(1):1–26. https://doi.org/10.1088/0034-4885/65/1/201

Tilman D (1999) Global environmental impacts of agricultural expansion: the need for sustainable and efficient practices. Proc Natl Acad Sci USA 96(11):5995–6000. https://doi.org/10.1073/pnas.96.11.5995

Ueyama M, Ichii K, Iwata H, Euskirchen ES, Zona D, Rocha AV, Harazono Y, Iwama C, Nakai T, Oechel WC (2013) Upscaling terrestrial carbon dioxide fluxes in Alaska with satellite remote sensing and support vector regression. J Geophys Research: Biogeosciences 118(3):1266–1281. https://doi.org/10.1002/jgrg.20095

United Nations (2023) SDGs Report 2023. In *The Sustainable development Goals Report 2023: Special Edition*. https://unstats.un.org/sdgs/report/2023/

Wade CM, Austin KG, Cajka J, Lapidus D, Everett KH, Galperin D, Maynard R, Sobel A (2020) What is threatening forests in protected areas? A global assessment of deforestation in protected areas, 2001–2018. Forests 11(5). https://doi.org/10.3390/F11050539

Wang W, Zhu Q, He G, Liu X, Peng W, Cai Y (2023) Impacts of climate change on pine wilt disease outbreaks and associated carbon stock losses. Agric for Meteorol 334(March):109426. https://doi.org/10.1016/j.agrformet.2023.109426

Wang B, Smith B, Waters C, Feng P, Liu DL (2024) Modelling changes in vegetation productivity and carbon balance under future climate scenarios in southeastern Australia. Sci Total Environ 924(December 2023):171748. https://doi.org/10.1016/j.scitotenv.2024.171748

Wei S, Yi C, Fang W, Hendrey G (2017) A global study of GPP focusing on light-use efficiency in a random forest regression model. Ecosphere 8(5). https://doi.org/10.1002/ecs2.1724

Willmott CJ, Rowe CM, Mintz Y (1985) Climatology of the terrestrial seasonal water cycle. J Climatol 5(6):589–606. https://doi.org/10.1002/joc.3370050602

Wouters H (2021) Downscaled bioclimatic indicators for selected regions from 1979 to 2018 derived from reanalysis. Copernicus Clim Change Service (C3S) Clim Data Store (CDS). https://doi.org/10.24381/cds.fe90a594

Wu Y, Wang P, Liu X, Chen J, Song M (2020) Analysis of regional carbon allocation and carbon trading based on net primary productivity in China. China Econ Rev 60(March 2019):101401. https://doi.org/10.1016/j.chieco.2019.101401

Xiao J, Zhuang Q, Baldocchi DD, Law BE, Richardson AD, Chen J, Oren R, Starr G, Noormets A, Ma S, Verma SB, Wharton S, Wofsy SC, Bolstad PV, Burns SP, Cook DR, Curtis PS, Drake BG, Falk M, Torn MS (2008) Estimation of net ecosystem carbon exchange for the conterminous United States by combining MODIS and AmeriFlux data. Agric for Meteorol 148(11):1827–1847. https://doi.org/10.1016/j.agrformet.2008.06.015

Xu L, Xu Y, Duan J, Wang Y, Yang H (2024) Assessing the spatial occupation and ecological impact of human activities in Chengguan district, Lhasa city, Tibetan Plateau. Sci Rep 14(1):1–14. https://doi.org/10.1038/s41598-024-57221-9

Yang F, Ichii K, White MA, Hashimoto H, Michaelis AR, Votava P, Zhu AX, Huete A, Running SW, Nemani RR (2007) Developing a continental-scale measure of gross primary production by combining MODIS and AmeriFlux data through support Vector Machine approach. Remote Sens Environ 110(1):109–122. https://doi.org/10.1016/j.rse.2007.02.016

Zhang H, Zhang Z (1999) Feedforward networks with monotone constraints. Proc Int Joint Conf Neural Networks 3:1820–1823. https://doi.org/10.1109/ijcnn.1999.832655

Zhang L, Wylie B, Loveland T, Fosnight E, Tieszen LL, Ji L, Gilmanov T (2007) Evaluation and comparison of gross primary production estimates for the Northern Great Plains grasslands. Remote Sens Environ 106(2):173–189. https://doi.org/10.1016/j.rse.2006.08.012

Zhang W, Luo G, Yuan X, Li C, Xie M, Wang Y, Ma X, Shi H, Hamdi R, Hellwich O, Ma X, Termonia P, De Maeyer P (2023) New data-driven method for estimation of net ecosystem carbon exchange at meteorological stations effectively increases the global carbon flux data. Methods Ecol Evol 14(9):2449–2463. https://doi.org/10.1111/2041-210X.14188

Zhong W, Haigh JD (2013) The greenhouse effect and carbon dioxide. Weather 68(4):100–105. https://doi.org/10.1002/wea.2072

Zhong Y, Xue Z, Davis CC, Moreno-Mateos D, Jiang M, Liu B, Wang G (2022) Shrinking habitats and native species loss under Climate Change: A Multifactorial Risk Assessment of China's Inland wetlands. Earth's Future 10(6). https://doi.org/10.1029/2021EF002630

Zhu W, Xie Z, Zhao C, Zheng Z, Qiao K, Peng D, Fu YH (2024) Remote sensing of terrestrial gross primary productivity: a review of advances in theoretical foundation, key parameters and methods. GIScience Remote Sens 61(1). https://doi.org/10.1080/15481603.2024.2318846

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Pedro Lago-González Pedro Lago-González (he/him), predoctoral researcher at the HydroForestry Geomodeling Research Group in the University of Vigo. Graduate in Environmental Science with a mention in climate and atmospheric processes by the University of Vigo (2021). Master of Science in Geophysics and Meteorology with specialization in meteorology by the University of Granada (2022). Currently working on his doctoral research, in which he studies the coupling mechanisms between the atmosphere and the biosphere in forest ecosystems.



Carolina Acuña Alonso Carolina Acuña-Alonso (she/her), researcher at the Applied Geotechnologies (GeoTech) Research Group at the Research Center in Technologies, Energy and Industrial Processes (CINTECX) of the University of Vigo and postdoc at the Centre for the Research and Technology of Agro-Environmental and Biological Sciences (CITAB) of the University of Trás-os-Montes e Alto Douro. PhD with international mention by the University of Vigo (2023). Graduate in Forestry Engineering by the University of Vigo (2018) and Master of Science in Environmental Engineering (2020) by the University of Santiago de Compostela. Her research scope focuses on the applications of remote sensing and artificial intelligence for the analysis of freshwater and land ecosystems.



Xana Álvarez Xana Álvarez (she/her), researcher at the HydroForestry Geomodeling Research Group in the University of Vigo. PhD with international mention by the University of Vigo (2015). Graduate in Forestry Engineering by the University of Vigo (2014) and Master of Science in Environmental Technologies (2011) and Technical Forest Engineering (2009), both by the University of Vigo. Her current research lines focus on the applications of remote sensing and artificial intelligence for the analysis of freshwater and land ecosystems.